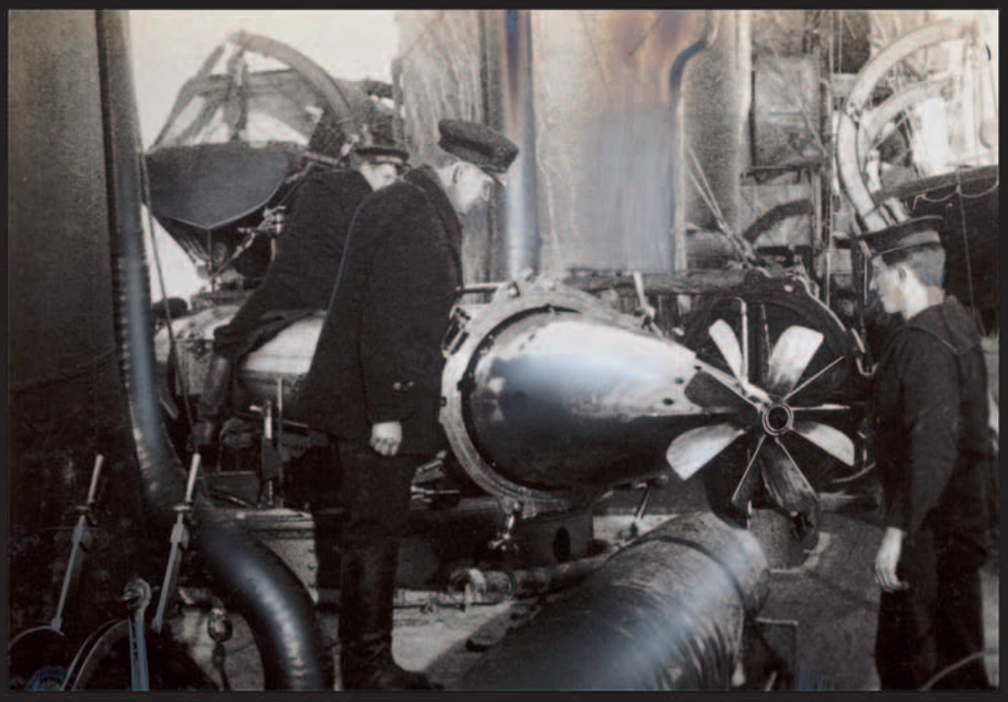
SMARTENING UP

Crews changed as well as ships. The press-gang disappeared and was replaced either by the signing up of volunteers for long-term service, or short-term conscription as part of a wider system of universal military service. It became standard practice for navies to have uniforms for their men. The old idea of training officers and men through practical experience at sea from boyhood died out. The United States set up the Naval Academy at Annapolis in 1850; the Japanese established a cadet academy at Etajima; and even the tradition-bound British felt obliged to create a Royal Naval College at Dartmouth in the early 20th century. Special training ships came into use for ratings. Men were still taken on board at a young age and given a hard time, with lots of caning and flicking with ropes. In general, naval punishment remained harsh, although milder than in the 18th century. The German navy was atypical in still flogging its conscript ratings in 1914. But discipline in some ways became far more rigorous than before, with meticulous regard for details of personal appearance and the cleanliness and painting of ships. Harassment over such relatively minor matters, along with complaints over pay and conditions, caused occasional outbreaks of mutiny below decks. Officers in general tended to become more remote from the men than in Nelson's day. Victorian and Edwardian navies were acutely class-conscious and promotion from the ranks was virtually non-existent.

STEAMING TO WAR

Coal-fired engines meant a new breed of engineers and stokers appeared on board, keeping the machinery going in often difficult and unpleasant conditions deep inside the ship. Armor plating left even less chance for air or natural light below decks than previously, although the introduction of electricity was a boon. Fire was less of a hazard than on wooden ships and there were no falls from rigging. But there were disturbing losses of warships to explosions, a consequence of problems with coal stocks or boilers, or with the new propellants and high explosives that sometimes proved unstable. Taking on coal was back-breaking manual labour, and the need for it limited ships' tactical flexibility, requiring breaks from action to replenish fuel stocks from coaling ships or to put in at coaling stations. In war, engagements were fought at ever increasing range but lost none of their savagery. New munitions produced new wounds, especially flash burns that proved a major source of casualties in World War I. Officers could no longer afford to stand exposed on an open deck swept by the blast of explosions and shards of steel. The old wooden ships had been hard to sink in battle, however much punishment they took, but a steel ship might go to the bottom in minutes, especially if a magazine was hit, taking most of the crew with her.



Inspecting a torpedo
These British sailors are inspecting a self-propelled torpedo. Whether deployed on small torpedo boats or submarines, self-propelled torpedoes challenged the dominance of big-gun battleships.





